Modern Cryptography and AI/ML Security

LECTURE 32

Visual Cryptography

IISER THIRUVANANTHAPURAM

Course ID: DSC 4110/6111 **Instructor:** Dr. Laltu Sardar **Date:** November 2, 2026

1 Visual Cryptography

Visual Cryptography (VC) is a secret-sharing technique that allows a secret image to be distributed among multiple participants in such a way that the original image can be reconstructed by stacking their shares. The reconstruction does not require any cryptographic computation or decryption algorithm. Instead, the secret becomes visually recognizable through the superposition of the shares.

2 Visual Cryptography: 2-out-of-2 Secret Image Sharing

In this section, we consider a basic **2-out-of-2 visual cryptography scheme**, where two participants hold two different shares of a secret black-and-white image. Both shares are required to reconstruct the secret image.

2.1 Pixel Expansion

Consider a binary image consisting of black and white pixels. We represent a black pixel by 1 and a white pixel by 0. In our construction, each pixel of the original image is expanded into a 2×2 block of four subpixels in each share. Let the corresponding blocks in the two shares be

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}.$$

The entries represent the colors of the subpixels, where 1 denotes black and 0 denotes white. The sharing procedure depends on whether the original pixel is black or white.

2.2 Sharing a Black Pixel

If the original pixel is black, we randomly select exactly two positions among the four subpixels in the first share and make them black. The remaining two subpixels are white.

For the second share, we select the complementary two positions and make them black.

For example, consider the following two blocks:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

When the two shares are stacked, the resulting block is

$$A \vee B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Here, $\vee$ denotes the Boolean OR operation, corresponding to the physical superposition of the two transparencies.

Thus, all four subpixels become black, and no light passes through the reconstructed block.

2.3 Sharing a White Pixel

If the original pixel is white, we again randomly select exactly two positions among the four subpixels in the first share and make them black. However, in this case, we select the **same positions** in the second share and make them black. For example,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

When the two shares are stacked, we obtain

$$A \vee B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Only two out of the four subpixels become black, while the remaining two stay white. Therefore, light can pass through the reconstructed block, making it appear lighter than the block corresponding to a black pixel.

Algorithm 1: 2-out-of-2 Visual Cryptography

Input: Binary secret image I of size $M \times N$
Output: Two shares S_1 and S_2

```
Initialize S_1, S_2 of size 2M × 2N;
foreach pixel I(i, j) do
    Randomly select two positions among four subpixels;
    Construct a 2 × 2 block A with 1 at the selected positions and 0 elsewhere;
    if I(i, j) = 1 then
        // Black pixel
        Construct B using the complementary positions;
    else
        // White pixel
        Set B = A;
    Place A in S_1 and B in S_2 at the corresponding expanded pixel position;
return S_1, S_2;
```

2.4 Reconstruction of the Secret Image

The secret image is reconstructed by physically stacking share A over share B, relying on light transmission through overlapping transparencies.

- **Black pixel:** Complementary black positions produce four black subpixels, blocking light.
- **White pixel:** Identical black positions produce two black and two white subpixels, allowing partial light transmission.

The resulting contrast enables visual recovery of the secret image without computational decryption.

2.5 Security Intuition

An individual share does not reveal information about the secret pixel. Each 2×2 block contains exactly two black and two white subpixels, whose positions are selected uniformly at random. Thus, the distribution of a share is identical whether the original pixel is black or white.

However, when both shares are stacked, the relationship between their black subpixel positions reveals the original pixel value.

Thus, the scheme provides the following properties:

1. **Perfect secrecy of individual shares:** A single share does not reveal information about the corresponding secret pixel.

2. **Visual reconstruction:** The secret image can be reconstructed by stacking both shares without requiring any computational decryption.

Key Observation. The security of 2-out-of-2 visual cryptography relies on the random selection of black subpixel positions. Although the individual shares look statistically identical for black and white secret pixels, stacking the two shares produces different patterns, enabling visual recovery of the secret.

Appendix

A Steganography Using Image Superposition (Not for exam)

Steganography is the technique of hiding secret information within an ordinary-looking object so that the existence of the secret itself is not easily detected.

In image-based steganography, the objective is to construct two or more images that individually appear meaningful or innocuous to an observer. However, when these images are overlapped, a hidden secret image becomes visible.

Unlike conventional visual cryptography, where individual shares need not have meaningful visual content, this approach aims to conceal the existence of the secret by making each share appear to be an ordinary image.

A.1 Construction

Consider two binary images I_1 and I_2, each of size $M \times N$. We use 1 to represent black and 0 to represent white.

Let S be the secret binary image. We construct two images I_1 and I_2 such that their physical superposition reveals S.

For each pixel position (i, j), define the images as follows:

$$I_1(i,j) = R(i,j),$$

$$I_2(i,j) = \begin{cases} 1 - R(i,j), & \text{if } S(i,j) = 1, \\ R(i,j), & \text{if } S(i,j) = 0, \end{cases}$$

where R is a randomly generated binary image.

The reconstruction is performed by physically overlapping the two images, corresponding to the Boolean OR operation:

$$S'(i,j) = I_1(i,j) \vee I_2(i,j).$$

Thus,

$$S'(i,j) = \begin{cases} 1, & \text{if } S(i,j) = 1, \\ R(i,j), & \text{if } S(i,j) = 0. \end{cases}$$

For a black secret pixel, the corresponding pixels in the two images are complementary, producing black after superposition. For a white secret pixel, the corresponding pixels are identical, preserving the randomly selected pixel value.

Consequently, the secret image can be recovered through the superposition of the two images.